DASARI VENKATA PANAKALA CHITTI BABU

Embedded Systems Engineer | FPGA, VLSI & AI-Driven Hardware
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SUMMARY

Electronics and Communication Engineering undergraduate (3rd year) with hands-on experience building embedded systems on ESP32, including a real-time health monitoring system with wireless data transmission and a CNN-based hand sign detection pipeline using computer vision. Familiar with FPGA fundamentals, Verilog HDL, and hardware–software co-design, with additional experience in full-stack web development (Next.js, TypeScript). Actively seeking an Embedded Systems / FPGA / VLSI / AIML Hardware Internship to apply practical engineering skills to real-world projects.

TECHNICAL SKILLS

- **Embedded Systems & Hardware:** ESP32, Microcontrollers, GPIO, UART, I2C, SPI, Sensor Interfacing, Real-Time Data Acquisition, Hardware Debugging
- **FPGA & VLSI:** Verilog HDL, Digital Logic Design, FPGA Fundamentals, Intel Quartus Prime Lite
- **Programming Languages:** Python, C, Embedded C
- **AI / Computer Vision:** OpenCV, MediaPipe, CNN, Machine Learning
- **Tools & Technologies:** MATLAB, Git, GitHub, VS Code, Jupyter Notebook
- **Web Development:** Next.js, TypeScript, Tailwind CSS, React, GSAP, Framer Motion

PROJECTS

AI-Powered Road Hazard Intelligence | IMU + Computer Vision + GPS Fusion *eMudhra Hackathon 2026*

- Designed a neuro-symbolic road hazard detection system fusing IMU vibration analysis, CNN-based visual classification (YOLO/MobileNet), and GPS-based multi-vehicle clustering
- Built a rule-based confidence-cascade escalation model (log → warn → alert) to convert multi-vehicle detections into driver warnings and authority alerts
- Conducted a literature survey across IEEE Xplore, arXiv, and MDPI to validate technical feasibility and novelty
- Produced a cost-estimated hardware architecture (ESP32, MPU6050, NEO-6M GPS, ESP32-CAM) for a 3-vehicle demo

Project Report

ESP32-Based Health Monitoring System | ESP32, Embedded C, Sensors *Sep 2025 – Oct 2025*

- Designed and implemented a real-time health monitoring system using ESP32, interfacing multiple physiological sensors for live vital parameter acquisition
- Enabled wireless data transmission for remote, real-time monitoring applications
- Performed end-to-end hardware–software integration, testing, and debugging to ensure stable system performance

Project Report for Health Moniter

Hand Sign / Word Sign Detection System | Python, OpenCV, MediaPipe, CNN *Nov 2025 – Dec 2025*

- Developed a computer vision pipeline using MediaPipe for hand landmark extraction from video input
- Built a CNN-based model for gesture/word-sign classification from extracted hand landmarks

Freelance Front-End Developer — APSLOCK | Next.js, TypeScript, Tailwind CSS *Mar 2026 – Apr 2026*

- Collaborated with a team to build the agency's client-facing website using Next.js, TypeScript, and Tailwind CSS
- Implemented custom animations and interactive UI components using GSAP and Framer Motion

https://www.apslock.com/

EDUCATION **Graduation: 2027**

Bachelor of Technology — Electronics & Communication Engineering

M. S. Ramaiah University of Applied Sciences, Bangalore, India

Expected Graduation: 2027 | Current Status: 3rd Year

Relevant Coursework: VLSI Design, Embedded Systems, Digital Electronics, C Programming, Python Programming

LEADERSHIP & EXTRACURRICULAR

Director of Club Services, Rotaract Club of Bangalore Warriors — Co-organized technical and social initiatives; strengthened teamwork, communication, and leadership skills through cross-functional collaboration.